Programme 2: Family Patterns

1. What are the sex cells that the body produces called?
2. What do the cells form in females? Males?
3. What is this particular secondary cell division of the sex cells called?
4. How many chromosomes do most human body cells contain?
5. How many chromosomes do the gametes of each sex cell contain?
6. Why is fertilisation necessary?
7. If chromosomes combine from the same two parents, why do children in a family show all kinds of genetic differences? And why are some male and some female?
8. Name at least one recessive and one dominant condition.
9. Match the following processes to mitosis or meiosis:

Occurs in body cells _____	**Chromosomes in cell duplicate** _____	**Identical chromosome sets separate.** _____
Cell contents duplicate _____	**Cell divides to form 2 cells of equal size.** _____	**This process happens in body cell of plants & animals.** _____
Occurs in sex cells _____	**Chromosomes duplicate** _____	**Form two identical pairs of chromotids** _____
Centromere is their joining point _____	**4 copies of each chromosome in the cell** _____	**Nucleus membrane disappears** _____
Spindle fibres grow across the cell _____	**Chromosome pairs line up with spindle fibres attached to centromeres** _____	**Spindle fibres contract, separating members of each pair** _____
Each chromosome still has 2 chromatids _____	**Division occurs again. Each cell now has half chromosomes of original cell** _____	**These cells are either sperm or egg** _____